Solution:

First, resolve the given number into prime factors. Group the identical factors in pairs and then take one from them to find the square root.

Now, $324 = 2\times2\times3\times3\times3\times3$

$= 2^2\times3^2\times3^2$

$= (2\times3\times3)^2$

$\therefore \sqrt{324} = \sqrt{(2\times3\times3)^2}$

$= 2\times3\times3$

$\therefore \sqrt{324} = 18$

2	324
2	162
3	81
3	27
3	9
3	3
	1

Example 1.23

Find the least number by which 250 is to be multiplied (or) divided so that the resulting number is a perfect square. Also, find the square root in that case.

Solution:

Here, $250 = 5\times5\times5\times2$

$= 5^2\times5\times2$

Here, the prime factors 5 and 2 do not have pairs.

5	250
5	50
5	10
2	2
	1

Therefore, we can either divide 250 by 10 (5×2) or multiply 250 by 10.

(i) If we multiply 250 by 10, we get $2500 = 5^2\times5\times2\times5\times2$ and therefore the square root of 2500 would be $5\times5\times2 = 50$.

(ii) If we divide 250 by 10, we get 25 and in this case we get $\sqrt{25} = \sqrt{5^2} = 5$.

Example 1.24

Is 108 a perfect square number?

Solution:

Here, $108 = 2\times2\times3\times3\times3$

$= 2^2\times3^2\times3$

Here, the prime factor 3 does not have a second pair. Hence, 108 is not a perfect square number.

2	108
2	54
3	27
3	9
3	3
	1



Think

In this case, if we want to find the smallest factor with which we can multiply or divide 108 to get a square number, what should we do?

1.7.2 Finding the square root of a number by Long Division Method

When we come across numbers with large number of digits, finding their square roots by factorisation becomes lengthy and difficult. Use of long division helps us in such cases. Let us look into the method with a couple of illustrations.

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